BINF*6970: Statistical Bioinformatics

Assignment 3

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Group #: 2

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Abstract

This study analyzes genotype data from the 1000 Genomes Project to classify individuals into East Asian (EAS) and European (EUR) populations using SNPs from ancestry-informative genes (SLC24A5 and EDAR). Principal Component Analysis (PCA) was used for dimensionality reduction and visualization, while Elastic Net logistic regression and Random Forest were used for classification. Results showed clear separation between populations in PCA space, across classification models, the SNP rs1426654 in the *SLC24A5* gene consistently emerged as the strongest predictor, aligning with its well-established role in population differentiation. In contrast, different SNPs within the *EDAR* gene were identified depending on the modeling approach: Elastic Net logistic regression selected a variant at 2:109552935_CG/C, while Random Forest identified rs922452.Random Forest achieved near-perfect classification performance, whereas Elastic Net provided interpretable feature selection through coefficient shrinkage. These findings highlight both the predictive power of ancestry-informative SNPs and the complementary strengths of the applied statistical models and high classification accuracy.

Background and Objective

There is a substantial genetic differentiation in human populations across various regions which is influenced by geographic location, ancestral migration and natural selection (Byrska-Bishop et al., 2022) . These variations are primarily driven by differences in DNA sequences, with single nucleotide polymorphisms (SNPs) being the most common form of variation. SNPs represent single base-pair changes in the genome and occur approximately once every 300 nucleotides, making them valuable markers for studying genetic diversity and population structure (Thorisson et al., 2005). Large-scale projects such as the 1000 Genomes Project have provided comprehensive catalogs of human genetic variation across diverse populations, enabling researchers to study ancestry and population differentiation at a genomic level (Genomes Project, 2015).

Two loci are among the most strongly differentiated between European (EUR) and East Asian (EAS) populations globally: *SLC24A5* and *EDAR*. The *SLC24A5* gene (chromosome 15q21.1) encodes a cation exchanger with a central role in melanosome biogenesis and skin pigmentation while *EDAR* gene (chromosome 2q13) encodes a TNF receptor family member critical for ectodermal development, influencing hair morphology, sweat gland density, incisor shape, and mammary gland ductal branching (Pavan & Sturm, 2019; Riddell et al., 2020). The nonsynonymous SNP rs1426654 within this gene accounts for a substantial proportion of the pigmentation difference between European and African populations, and is present at near-fixation frequency in Europeans but not in East Asians (Basu Mallick et al., 2013). Also, for the EDAR, the derived V370A allele at rs3827760 reached near-fixation in East Asian and Native American populations as a consequence of ancient positive selection, and is essentially absent in European populations, making it an ideal ancestral informative marker in East Asian classification (Kimura et al., 2009).

With all these ancestral linkages in sight, the objective of this study is to classify individuals into European (EUR) and East Asian (EAS) populations using SNP data from ancestry informative genes (SLC24A5 and EDAR) gotten from the 1000 Genomes Project dataset as well as comparing the performance of different statistical learning methods mainly logistic regression and cross-validation and random forests.

Details of the Dataset

Genotype data were obtained from the 1000 Genomes Project Phase 3 release, accessed via the International Genome Sample Resource (IGSR) FTP server. The Phase 3 resource provides whole-genome sequence data from 2,504 unrelated individuals across five super-populations; African (AFR), Admixed American (AMR), East Asian (EAS), European (EUR), and South Asian (SAS). We retained just two super populations which further reduced our data to 1007; 504 EAS and 503 EUR. The genotype data were extracted for two target loci of the genes *SLC24A5* and *EDAR* from GRCh37 assembly and the SNP data (575 SNPs for SLC24A5 and 3482 SNPs for EDAR) was gotten from the IGSR data slicer which after filtering resulted to a combined 454 SNPs which was subjected form the analysis. Population metadata were retrieved from the IGSR panel file (integrated_call_samples_v3.20130502.ALL.panel).

Raw genotype calls in VCF format were parsed using the *VariantAnnotation* R package and encoded numerically as allele dosage; Homozygous major allele 0, Heterozygous 1, Homozygous minor allele 2. Quality control proceeded in three stages: removal of SNPs with any missing genotype data (NAs), exclusion of SNPs with zero variance and low minor allele frequency filtering retaining only SNPs with Minor Allele Frequency > 0.01. After all these quality control steps, the SNPs (454) was used for the analysis and PCA was conducted on the filtered genotype to check for clear separation or clustering.

Methods

Logistic Regression Methods

Genotype and ethnography data was split 75/25 to training and test sets. A series of Elastic Net models were tested, with the best alpha and lambda values for each model (by end result accuracy) obtained through an alpha search grid at intervals of 0.01. Accuracy was measured using deviance.

A total of four Elastic-Net logistic regression models were produced, comparing each combination of 10-fold and 20-fold cross-validation methods, with fitting by either minimum cross-validation error (lambda.min), or fitted with a cross-validation error 1 standard error from the minimum (lambda.1se). As principal component analysis (PCA) indicated a moderate proportion of the data lay unaccounted for by the first two PCs (~28%), the top 11 predictors were investigated to check for absolute effect trends, and the total number of non-zero coefficients was checked for each set. This method was used due to its effectiveness in classifying categorical binary variables. This was also used to study the variations seen from the PCA analysis.

Random Forest

Genetic Variation (SNP) and population(ethnography) data were split into 70/30 training and test sets respectively. A random forest classification model was trained using repeated K-fold cross validation with 5 folds and 3 repeats for accurate model performance. Hyperparameter tuning focused on the number of variables randomly sampled at each split (mtry), which was systematically evaluated over a predefined grid of values (10, 20, 30, and 50). The optimal model was selected based on cross-validated performance metrics obtained during training. The error was estimated using the caret package to compare the prediction accuracy of the model.

This method was employed to handle high dimensional, complex data and to account for non linear interactions within the SNP data.

Results Summary

Principal Component Analysis

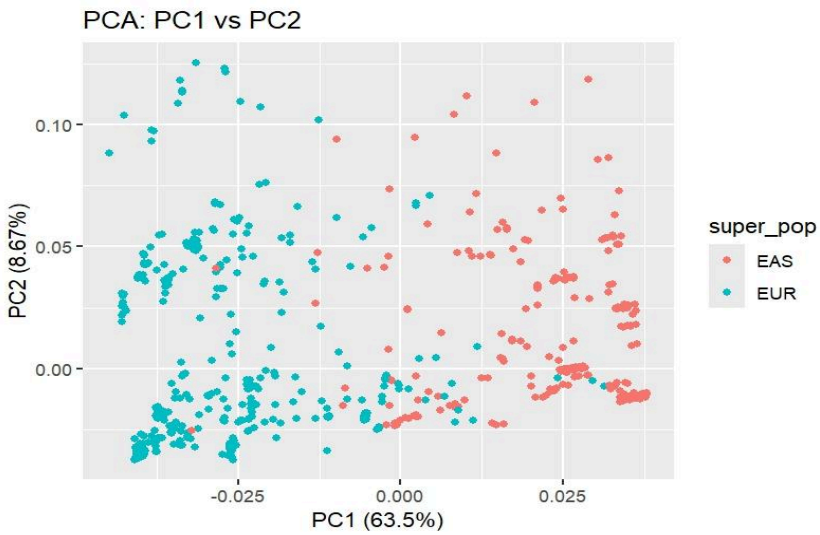


Figure 1: Principal Component Analysis
Principal component analysis of genetic variation showing clear separation between European (EUR) and East Asian (EAS) populations. PC1 explains 63.5% of the variance and primarily distinguishes the two groups, while PC2 (8.67%) captures additional within-population variation.

This was used to understand the structure of the data and check if there are observable clusters before further analysis.

Logistic Regression Results

Comparison of logistic regression models identifying SNPs among the chromosome 15 SLC24A5 ("15:") gene and the chromosome 2 EDAR ("2:") gene to predict ethnography status between East Asian and European superpopulations:

Tables #1-4: Comparison of Elastic Net logistic regression models.

- For min CV models:

```
> print(m10_top_min)
                                Predictor Coefficient Absolute_Effect Rank
`15:48426484_A/G`         `15:48426484_A/G`  -2.2253209       2.2253209    1
`15:48433494_A/G`         `15:48433494_A/G`  -1.3493208       1.3493208    2
`2:109552935_CG/C`       `2:109552935_CG/C`   0.5557519       0.5557519    3
`15:48415765_T/C`         `15:48415765_T/C`  -0.5536027       0.5536027    4
`2:109513601_A/G`         `2:109513601_A/G`  -0.4998115       0.4998115    5

> print(m20_top_min)
                                Predictor Coefficient Absolute_Effect Rank
`15:48426484_A/G`         `15:48426484_A/G`  -2.4462869       2.4462869    1
`15:48433494_A/G`         `15:48433494_A/G`  -1.4386317       1.4386317    2
`15:48415765_T/C`         `15:48415765_T/C`  -0.5916232       0.5916232    3
`2:109552935_CG/C`       `2:109552935_CG/C`   0.5902390       0.5902390    4
`2:109513601_A/G`         `2:109513601_A/G`  -0.5244031       0.5244031    5
```

- For 1se models:

```
> print(m10_top_lse)
                           Predictor Coefficient Absolute_Effect Rank
`15:48426484_A/G`   `15:48426484_A/G`  -1.7348658       1.7348658    1
`15:48433494_A/G`   `15:48433494_A/G`  -1.1761200       1.1761200    2
`15:48415765_T/C`   `15:48415765_T/C`  -0.4748444       0.4748444    3
```

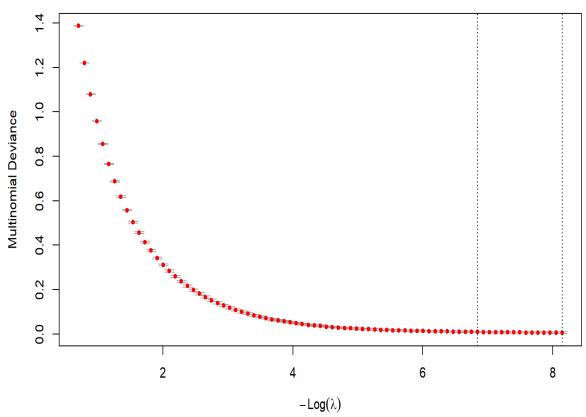
```
`2:109552935_CG/C` `2:109552935_CG/C`   0.4636577       0.4636577    4
`2:109513601_A/G`   `2:109513601_A/G`  -0.4329563       0.4329563    5

> print(m20_top_lse)
                                 Predictor Coefficient Absolute_Effect Rank
`15:48426484_A/G`          `15:48426484_A/G`  -2.2935440       2.2935440    1
`15:48433494_A/G`          `15:48433494_A/G`  -1.3111873       1.3111873    2
`2:109552935_CG/C`        `2:109552935_CG/C`   0.5659937       0.5659937    3
`15:48415765_T/C`          `15:48415765_T/C`  -0.5241609       0.5241609    4
`2:109513601_A/G`          `2:109513601_A/G`  -0.5008964       0.5008964    5
```

"m10" and "m20" indicate the model was fitted using 10 fold or 20 fold cross validation (respectively). "min" indicates the model was fitted using the minimum cross-validation error. "1se" indicates the model was fitted with a cross-validation error 1 standard deviation from the minimum.

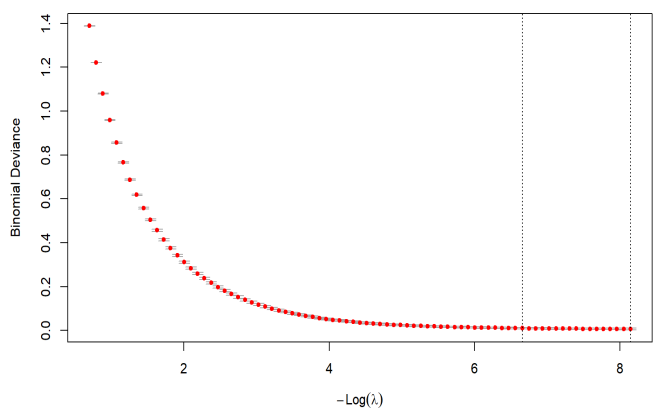
The top 5 predictors remain consistent across all models, with the SNPs '15:48426484_A/G' and '15:48433494_A/G' as first and second most impactful predictors. The SNPs '2:109552935_CG/C', '15:48415765_T/C', and '2:109543883_C/T' remain ranked third, fourth and fifth most impactful predictors across all models, though their ordering within that ranking shifts.

Model Generation and Accuracy Comparison Figures:

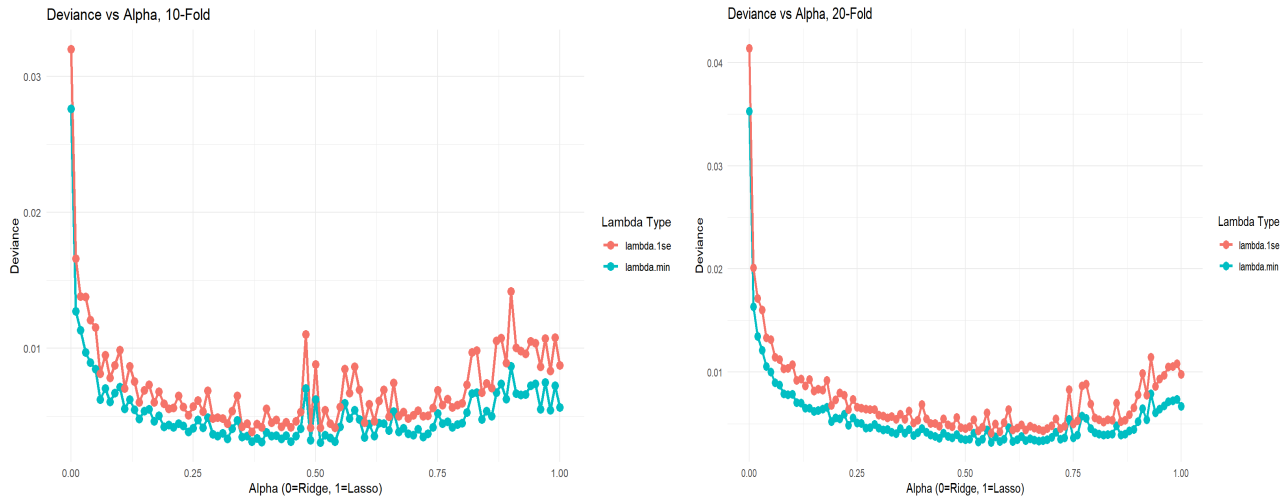
Elastic Net (10-Fold) - Deviance:



Elastic Net (20-Fold) - Deviance:



Figures 2-3: The leftmost dashed lines correspond to the log values of the λmin, while the rightmost dashed line corresponds to that of the λ1*se.* Typically, minimum cross validation (CV) error models tend to incorporate more coefficients into the model, while 1se models restrict coefficient count. Interestingly enough, this behaviour was not observed in the 10-fold cross validation models here, where the best 10-fold min CV model incorporated 87 non-zero coefficients, while the best 10-fold 1se model incorporated 120 non zeros.



Figures 4-5: Accuracy was measured using deviance. Alpha value impact on deviance was quite clear, with models gaining increasing accuracy with middling values. The trend is slightly more erratic in 10-fold models, but remains apparent. Min CV models remained consistently more accurate than 1se models

Although they had a higher deviance ceiling at low-alpha values, 20-fold models slightly outperformed 10 fold models by accuracy. This may be due to 10-fold models being slightly more vulnerable to overfitting, as more of the overall datapool is conserved in training rounds.

The best model overall by accuracy was a 20-fold cross validation using minimum CV error. This model had a reported alpha value of 0.56, and cross-validation deviance error (lambda.min) of approximately 0.00029. This model incorporated 83 non-zero coefficients.

Random Forest Results

After hyperparameter tuning and model training it was observed that the model produced the best results when 30 predictors were selected.

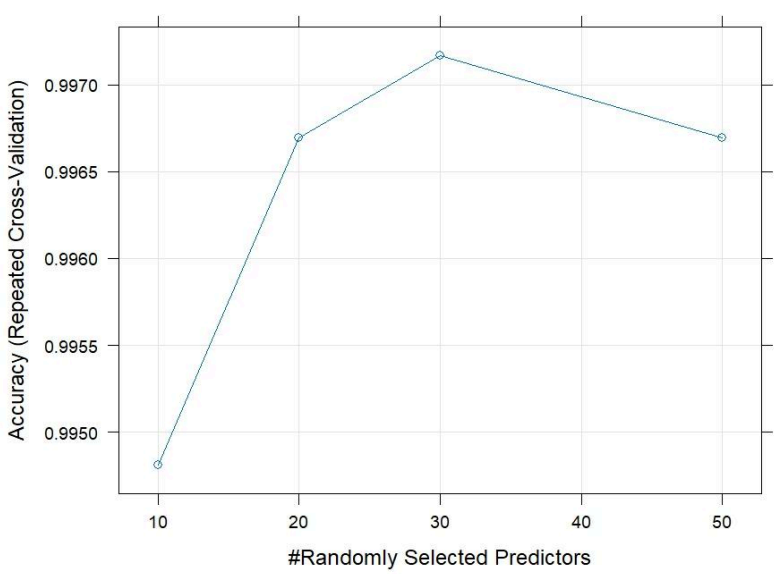


Figure 6: Hyperparameter Optimization for Random Forest
Relationship between the number of randomly selected predictors (mtry) and the model's predictive accuracy. The plot shows that optimal cross-validation accuracy is achieved when selecting 30 predictors, with performance beginning to plateau or decline thereafter.

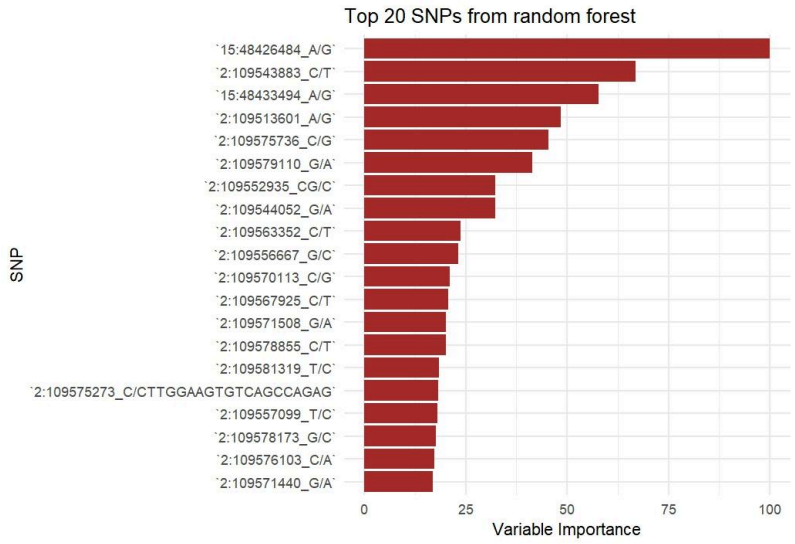


Figure 7: Top 20 Most Important SNPs
Feature importance ranking of the top 20 Single Nucleotide Polymorphisms (SNPs) as determined by the Random Forest algorithm. Importance is measured based on the total decrease in node impurities, highlighting the primary genetic variants contributing to the model's predictive performance.

General Method Comparison between logistic regression and Random Forest

S/N	Logistic Regression	Random Forest
1.	The best model had 83 predictors	Best Performance at 30 predictors
2.	Involves a lot of computation	It is less computationally intensive
3.	Cross validation error of 0.0003	Test Error of 0.003
4.	Top 10 SNPs had 3 SNPs from SLC24A5	Top 10 SNPs had 2 SNPs from SLC24A5
5.	Top EDAR SNP was 2:109552935_CG/C	Top EDAR SNP was 2:109543883_C/T

Similarities

1. Both methods were able to accurately predict population from the SNP data
2. Both methods arrived at the most important SNP being rs1426654 SNP (15:48426484_A/G) of the SLC24A5 gene.
3. Both methods used cross validation.

Discussion

This analysis used SNP data to classify populations into continental classes and to test which ones were strongest predictors based on logistic regression and random forest models. The rs1426654 SNP (15:48426484_A/G), of the SLC24A5 gene, was found to be the strongest predictor in all tested models, for both logistic regression and random forest methods. However, for the EDAR gene, two

different SNPs were determined as top predictors: the SNP located at 2:109552935_CG/C from logistic regression, and the rs922452 SNP (2:109543883_C/T) from the random forest (Jarve et al., 2019).

The rs1426654 SNP in SLC24A5 aligned greatly with prior literature, as many studies have highlighted it as a defining feature often differentiating African and especially Eurasian populations. In particular, the lighter-skinned A allele of this SNP (rs1426654-A) has been mostly instrumental in shaping further skin pigmentation diversity across South Asia, as evident by its differing frequencies throughout the Indian subcontinent (Mallick et al., 2013). In contrast, there has been stronger natural selection of this allele in Europe, leading to more fixed frequencies in the continent (Mallick et al., 2013). Based on studies conducted on the *golden* phenotype of said gene in zebrafish models, mostly analogous to the lighter A allele phenotype in humans, a possible explanation for this trait could be attributed to reduced melanosomal number or density, along with a more prolonged development period of melanin pigmentation (Lamason et al., 2005). On the other hand, the ancestral G allele (rs1426654-G), a more conserved phenotype closely linked to African populations, usually resulted in darker skin pigmentations due to lower pheomelanin (also responsible for lighter pigmentation) from less ion transport (Mpofana et al., 2025). Unfortunately, this ancestral phenotype has also exhibited possible connections to the disease melasma, most prominently in women of African (and Asian or Latin American) descent, potentially due to further hyperpigmentation of darker skin that may be influenced by environmental conditions, such as UV exposure (Mpofana et al., 2025).

Meanwhile, taking a closer look at the SNPs identified for EDAR, the SNPs revealed from each method led to variations in potentially two different morphological traits between populations. The first SNP (2:109552935_CG/C) within the EDAR region did not appear to be directly associated with a specifically known SNP, though the closest known SNP in proximity that could actually explain this variation was likely the rs3827760 SNP (SNPedia, n.d., Fondevila et al., 2013, Riddell et al., 2020). Having a higher prevalence in Asia and Native America, known for encoding Val370Ala (V370A) substitution in the EDAR gene, this SNP has been known for variation in dental morphology, including properties such as cusps and roots (Riddell et al., 2020, Kimura et al., 2009). The rs922452 SNP, in contrast, pointed more towards the trait of hair morphology in European and Asian populations (Fujimoto et al., 2007). Specifically, this SNP showed connections to the EDAR gene regarding variations in the thickness of hair, with the C allele diverging towards Asian populations, leading to greater hair thickness, as opposed to the T allele in Europe (Fujimoto et al., 2007). Assays revealed a possible explanation for this variation via a decrease in luciferase activity in the C allele, which appeared to further affect the nuclear factor of kappa-light-chain-enhancer of activated B (NF-kB) pathway downstream, and with its suppression, this seemed to stimulate more hair growth in Asian populations (Fujimoto et al., 2007). Altogether, this analysis helped to explore some of the key genes driving human diversity across populations, as well as to explain how variations in SNPs were able to distinguish different populations between the African and Eurasian geographical areas.

References:

The 1000 Genomes Project Consortium, Corresponding authors, Auton, A., Abecasis, G. R., Steering committee, Altshuler, D. M., Durbin, R. M., Abecasis, G. R., Bentley, D. R., Chakravarti, A., Clark, A. G., Donnelly, P., Eichler, E. E., Flicek, P., Gabriel, S. B., Gibbs, R. A., Green, E. D., Hurles, M. E., Knoppers, B. M., … Abecasis, G. R. (2015). A global reference for human genetic variation. Nature, 526(7571), 68–74. https://doi.org/10.1038/nature15393

Basu Mallick, C., Iliescu, F. M., Möls, M., Hill, S., Tamang, R., Chaubey, G., Goto, R., Ho, S. Y. W., Gallego Romero, I., Crivellaro, F., Hudjashov, G., Rai, N., Metspalu, M., Mascie-Taylor, C. G. N., Pitchappan, R., Singh, L., Mirazon-Lahr, M., Thangaraj, K., Villems, R., & Kivisild, T. (2013). The Light Skin Allele of SLC24A5 in South Asians and Europeans Shares Identity by Descent. PLoS Genetics, 9(11), e1003912. https://doi.org/10.1371/journal.pgen.1003912

Byrska-Bishop, M., Evani, U. S., Zhao, X., Basile, A. O., Abel, H. J., Regier, A. A., Corvelo, A., Clarke, W. E., Musunuri, R., Nagulapalli, K., Fairley, S., Runnels, A., Winterkorn, L., Lowy, E., Eichler, E. E., Korbel, J. O., Lee, C., Marschall, T., Devine, S. E., … Zody, M. C. (2022). High-coverage whole-genome sequencing of the expanded 1000 Genomes Project cohort including 602 trios. Cell, 185(18), 3426-3440.e19. https://doi.org/10.1016/j.cell.2022.08.004

Dimisianos, G., Stefanaki, I., Nicolaou, V., Sypsa, V., Antoniou, C., Poulou, M., Papadopoulos, O., Gogas, H., Kanavakis, E., Nicolaidou, E., Katsambas, A. D., & Stratigos, A. J. (2009). A study of a single variant allele (rs1426654) of the pigmentation-related gene SLC24A5 in Greek subjects. *Experimental Dermatology*, *18*(2), 175–177. https://doi.org/10.1111/J.1600-0625.2008.00758.X

Fujimoto, A., Kimura, R., Ohashi, J., Omi, K., Yuliwulandari, R., Batubara, L., Mustofa, M. S., Samakkarn, U., Settheetham-Ishida, W., Ishida, T., Morishita, Y., Furusawa, T., Nakazawa, M., Ohtsuka, R., & Tokunaga, K. (2008). A scan for genetic determinants of human hair morphology: EDAR is associated with Asian hair thickness. Human Molecular Genetics, 17(6), 835–843. https://doi.org/10.1093/hmg/ddm355

Giardina, E., Pietrangeli, I., Martínez-Labarga, C., Martone, C., Angelis, F. de, Spinella, A., Stefano, G. De, Rickards, O., & Novelli, G. (2008). Haplotypes in SLC24A5 Gene as Ancestry Informative Markers in Different Populations. *Current Genomics*, *9*(2), 110. https://doi.org/10.2174/138920208784139528

Järve, M., Saag, L., Scheib, C. L., Pathak, A. K., Montinaro, F., Pagani, L., Flores, R., Guellil, M., Saag, L., Tambets, K., Kushniarevich, A., Solnik, A., Varul, L., Zadnikov, S., Petrauskas, O., Avramenko, M., Magomedov, B., Didenko, S., Toshev, G., … Villems, R. (2019). Shifts in the Genetic Landscape of the Western Eurasian Steppe Associated with the Beginning and End of the Scythian Dominance. Current Biology, 29(14), 2430-2441.e10. https://doi.org/10.1016/j.cub.2019.06.019

Kimura, R., Yamaguchi, T., Takeda, M., Kondo, O., Toma, T., Haneji, K., Hanihara, T., Matsukusa, H., Kawamura, S., Maki, K., Osawa, M., Ishida, H., & Oota, H. (2009). A Common Variation in EDAR Is a Genetic Determinant of Shovel-Shaped Incisors. *American Journal of Human Genetics*, *85*(4), 528–535. https://doi.org/10.1016/j.ajhg.2009.09.006

Lamason, R. L., Mohideen, M.-A. P. K., Mest, J. R., Wong, A. C., Norton, H. L., Aros, M. C., Jurynec, M. J., Mao, X., Humphreville, V. R., Humbert, J. E., Sinha, S., Moore, J. L., Jagadeeswaran, P., Zhao, W., Ning, G., Makalowska, I., McKeigue, P. M., O'Donnell, D., Kittles, R., … Cheng, K. C. (2005). SLC24A5, a Putative Cation Exchanger, Affects Pigmentation in Zebrafish and Humans. Science, 310(5755), 1782–1786. https://doi.org/10.1126/science.1116238

Mpofana, N., Mlambo, Z. P., Makgobole, M. U., Dlova, N. C., & Naicker, T. (2025). Association of Genetic Polymorphisms in SLC45A2, TYR, HERC2, and SLC24A in African Women with Melasma: A Pilot Study. International Journal of Molecular Sciences, 26(3), 1158. https://doi.org/10.3390/ijms26031158

Park, J. H., Yamaguchi, T., Watanabe, C., Kawaguchi, A., Haneji, K., Takeda, M., Kim, Y. Il, Tomoyasu, Y., Watanabe, M., Oota, H., Hanihara, T., Ishida, H., Maki, K., Park, S. B., & Kimura, R. (2012). Effects of an Asian-specific nonsynonymous EDAR variant on multiple dental traits. *Journal of Human Genetics*, *57*(8), 508–514. https://doi.org/10.1038/JHG.2012.60

Pavan, W. J., & Sturm, R. A. (2019). The Genetics of Human Skin and Hair Pigmentation. Annual Review of Genomics and Human Genetics, 20(1), 41–72. https://doi.org/10.1146/annurev-genom-083118-015230

Riddell, J., Basu Mallick, C., Jacobs, G. S., Schoenebeck, J. J., & Headon, D. J. (2020). Characterisation of a second gain of function EDAR variant, encoding EDAR380R, in East Asia. *European Journal of Human Genetics*, *28*(12), 1694. https://doi.org/10.1038/S41431-020-0660-6

Thorisson, G. A., Smith, A. V., Krishnan, L., & Stein, L. D. (2005). The International HapMap Project Web site: Figure 1. *Genome Research*, *15*(11), 1592–1593. https://doi.org/10.1101/gr.4413105

rs3827760 - SNPedia. (n.d.). Retrieved April 2, 2026, from https://bots.snpedia.com/index.php/Rs3827760

Task List

1. VCF framework data loading and processing
2. PCA scatter plot visualization and labelling
3. Random Forest prediction code and plots
4. Logistic Regression code and model training
5. Elastic Net model selection and visualization
6. SNP prediction comparison
7. Abstract and Background
8. Logistic Regression Method Writeup
9. Random Forest Method Writeup and Comparison Table
10. SNP Discussion
11. References

S/N	Task	Iroayo (CD%)	Sodiq (CD%)	Cyrus (CD%)	William (CD%)
R Coding					
1.	VCF framework data loading and processing	100	-	-	-
2.	PCA scatter plot visualization and labelling	100	-	-	-
3.	Random Forest prediction code and plots	100	-	-	-
4.	Logistic Regression code and model training	-	-	-	100
5.	Elastic Net model selection and visualization	-	-	-	100
6.	SNP prediction comparison	-	-	-	100
Report					
7.	Abstract and Background	-	100	-	-
8.	Logistic Regression Method Writeup	-	-	-	100
9.	Random Forest Method Writeup and Comparison Table	100	-	-	-
10.	SNP Discussion	-	-	100	-
11.	References	25	25	25	25